

Covariate-Conditional EVT for Extreme VaR Forecasting

A Bitcoin Application

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Abstract

Estimating the extreme Value-at-Risk of volatile assets such as Bitcoin is essential for risk management, but challenging since data are sparse at extreme quantile levels. The established benchmark is GARCH-EVT, which conditions on past squared returns but only partially captures macroeconomic effects. Because research has shown that extreme Bitcoin tails are mainly driven by macroeconomic covariates, we apply the covariate-conditional EVT-QR estimators of Wang et al. (2012) (2Stage) and Wang and Li (2013) (3Stage), and compare their extreme VaR forecasts to those of GARCH-EVT.

We find that the two main failure modes of the 2Stage model are exponential nonlinearity and covariate-varying extreme value index (EVI), with a sensitivity analysis establishing that the magnitude of EVI variation matters most. The rolling window backtest on Bitcoin data indicates no significant differences in full-sample performance. However, regime-conditional analysis shows that GARCH-EVT outperforms in the middle regime. This advantage diminishes as the quantile level increases, which is in line with the increasing influence of macroeconomic drivers at more extreme tail levels. In the linear and nonlinear regimes, GARCH-EVT does not exhibit a significant advantage, and 3Stage outperforms 2Stage in the nonlinear regime. Nevertheless, the practical benefit of 3Stage is constrained by its finite-sample instability. Covariate-conditional EVT-QR attains performance comparable to GARCH-EVT under more linear or nonlinear market conditions, but does not yield a general improvement at the current sample size, pointing to the degrees-of-freedom limitations inherent in extreme quantile estimation.

KEYWORDS: Extreme Conditional Quantiles, Value-at-Risk, Tail Extrapolation, Bitcoin, Extreme Value Index

1 Introduction

Events such as high insurance claims and high medical costs are rare, but when they do occur, the consequences can be severe. When describing the distribution of these losses, simple summary statistics such as the mean and median are insufficient for providing a complete picture of such risk. In order to incorporate the risk of these events into risk management, we are required to estimate the extreme quantiles of the response variable. For light-tailed data, this is relatively straightforward, but for heavy-tailed data, observations are sparse in the extreme tail, and direct estimation becomes unreliable. This is especially difficult in financial risk management, since financial asset returns are well-known to exhibit heavy tails (McNeil et al., 2005), and standard risk models break down at extreme quantile levels such as $\tau \in \{0.99, 0.999\}$. A direct quantile regression method, such as the CAViaR (Engle and Manganelli, 2004), models the conditional quantile as a dynamic process, but only estimates directly at the requested level. At extreme quantile levels, the tail contains just a handful of observations, making direct estimation statistically infeasible.

Extreme value theory (EVT) (de Haan and Ferreira, 2006) provides a solution by establishing a theoretical basis for extrapolating beyond the observed data range. A standard implementation is the GARCH-EVT (McNeil and Frey, 2000) model, in which a volatility model is fitted, and EVT is applied to the standardised residuals. However, GARCH-EVT conditions exclusively on past squared returns to determine the volatility forecast, and for assets whose tail risk is driven by factors beyond past volatility, this is a meaningful restriction.

Among financial assets, Bitcoin is a particularly demanding case. Its returns have been documented to exhibit heavy tail behaviour (Chaim and Laurini, 2018), abrupt regime changes, and extreme events such as the COVID-19 crash (2020), the Terra/Luna collapse (2022), and the FTX bankruptcy (2022). Kwon (2021) and Lawuobahsumo et al. (2024) demonstrated that the drivers of Bitcoin tail risk are, at extreme quantile levels, largely macroeconomic and market-stress factors. Volatility filtering explains only part of this, which makes Bitcoin a natural and demanding test case for an approach which combines extrapolation beyond the data range with conditioning on external covariates.

Wang et al. (2012) achieve exactly this by estimating intermediate conditional quantiles via quantile regression and extrapolating to the far tail using EVT, providing asymptotically justified estimates where direct estimation is infeasible. Their estimator, however, assumes linearity of the conditional quantile function on the original scale, which fails when the EVI varies with covariates. Wang and Li (2013) correct for this loss of linearity via a Box-Cox power transformation, applied before quantile regression. This restores linearity on the transformed scale, enabling reliable estimation even when tail heaviness varies across covariates. We therefore adopt the 3Stage estimator of Wang and Li (2013) as our primary method, with the 2Stage estimator of Wang et al. (2012) as an intermediate benchmark isolating the marginal contribution of the power transformation, and GARCH-EVT (McNeil and Frey, 2000) as the industry-standard benchmark.

Comparing these estimators raises our main empirical question of whether the added covariate flexibility of the EVT-QR framework makes up for the absence of explicit volatility structure, and which market conditions favour each approach. In addressing this question, we make several contributions to the literature. In addition to replicating the 2Stage and 3Stage estimators of Wang et al. (2012) and Wang and Li (2013) in an empirical financial risk management setting, we perform

a simulation sensitivity analysis that shows the degree of EVI variation determines 2Stage's failure and that the tolerance threshold shifts with the quantile level. Beyond this, we evaluate estimator performance among regimes defined by the degree of nonlinearity in the conditional quantile function, finding that GARCH-EVT leads only in the middle regime, but loses its edge in both the linear and nonlinear regimes.

The remainder of this thesis is structured as follows. Section 2 presents the 2Stage and 3Stage estimators and their theoretical foundations, followed by the empirical evaluation framework, including the GARCH-EVT benchmark. Section 3 contains a simulation study evaluating the performance of the 2Stage and 3Stage estimators across a range of heavy-tailed data-generating processes. Section 4 presents the empirical application to Bitcoin in which we evaluate estimator performance via a rolling window backtest. Finally, Section 5 concludes.

2 Methodology

2.1 Covariate-Conditional EVT Estimators

2.1.1 Model Setup and Assumptions

Suppose we observe the random sample $\{(y_i, \mathbf{x}_i), i = 1, ..., n\}$ of the random vector $(Y, \mathbf{X})$. Here y_i is the univariate response variable of interest, and $\mathbf{x}_i = (1, \tilde{x}_i^\top)^\top$ is the p-dimensional covariate vector with leading intercept. Then let $Q_Y(\tau|\mathbf{x}) = \inf\{y : F_Y(y|\mathbf{x}) \geq \tau\}$ be the τth conditional quantile of Y given $X = \mathbf{x}$, where $F_Y(\cdot|\mathbf{x})$ is the cumulative conditional distribution of Y given $X = \mathbf{x}$. In this article, we will estimate extreme high conditional quantiles $Q_Y(\tau_n|\mathbf{x})$, where $\tau_n \to 1$ as $n \to \infty$ at any rate, focussing on heavy-tailed distributions. In practice, direct quantile estimates at extreme tails are often unstable because of sparse data in this area. This led to the consideration of the 2Stage and 3Stage estimators.

We assume throughout that the conditional distribution $F_Y(\cdot \mid \mathbf{x})$ belongs to the maximum domain of attraction (MDA) of an extreme value distribution, $F_Y(\cdot \mid \mathbf{x}) \in D(G_{\gamma(\mathbf{x})})$, where $\gamma(\mathbf{x})$ is the varying EVI determining tail heaviness. This assumption is standard in extreme value literature (de Haan and Ferreira, 2006) and covers a wide range of heavy-tailed distributions. This assumption implies that, for a given random sample $Z_1, \ldots, Z_n$ from $F_Y(\cdot \mid \mathbf{x})$, there exist normalising constants $a_n > 0$ and $b_n \in \mathbb{R}$ such that

$$P\left(\frac{\max_{1\leq i\leq n} Z_i - b_n}{a_n} \leq z\right) \to G_{\gamma(\mathbf{x})}(z) = \exp\left[-(1+\gamma(\mathbf{x})z)^{-1/\gamma(\mathbf{x})}\right] \tag{1}$$

as $n \to \infty$, for $1 + \gamma(\mathbf{x})z \geq 0$. Throughout, we restrict to the case $\gamma(\mathbf{x}) > 0$ for all $\mathbf{x}$, corresponding to the Fréchet domain of attraction, which covers heavy-tailed distributions such as the Pareto and Student-t (de Haan and Ferreira, 2006).

The asymptotic theory for the 2Stage and 3Stage estimators shows consistency and asymptotic normality of the intermediate quantile estimates, tail index estimator, and final Weissman extrapolation step, under a set of regularity conditions on the tail behaviour, quantile regression design, and the growth rate of k. The 3Stage estimator additionally requires the power transformation parameter λ (Section 2.1.3) to lie in a compact set and be uniquely identified. The full set of conditions is stated in Wang et al. (2012) (conditions A1–A2, B1–B4) and Wang and Li (2013) (conditions B1–B6),

and we assume these hold throughout.

2.1.2 Intermediate Quantile Estimation

We consider the following linear quantile regression model:

$$Q_Y(\tau \mid \mathbf{x}) = \alpha(\tau) + \tilde{\mathbf{x}}^\top \beta(\tau), \quad \text{for } \tau \in [\tau_c, 1] \tag{2}$$

where $\alpha(\tau)$ is the intercept and $\beta(\tau)$ is the vector of slope coefficients, both varying with τ. This specification is referred to as the Non-Common Slopes (NCS) estimator (Wang et al., 2012). In contrast, the Common Slopes (CS) estimator restricts $\beta(\tau) = \beta_c$ to be constant across quantile levels.

Following Wang and Li (2013), we define a sequence of intermediate quantile levels $\tau_c < \tau_{n-k} < \tau_{n-k+1} < \cdots < \tau_m \in (0,1)$, where $m = n - [n^\eta]$ with $\eta \in (0,1)$ a small constant, and $\tau_j = j/(n+1)$. We set $\eta = 0.1$ throughout, since Wang et al. (2012) state that values $\eta \in (0, 0.2)$ perform well. Then for each $j = n-k, \ldots, m$, define

$$(\hat{\alpha}(\tau_j), \hat{\beta}(\tau_j)) = \arg\min_{\alpha,\beta} \sum_{i=1}^{n} \rho_{\tau_j}(y_i - \alpha - x_i^\top \beta),$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}_{u<0})$ is the quantile loss function (Koenker, 2005), and let $\theta(\tau) := (\alpha(\tau), \beta(\tau)^\top)^\top$. For a given $\mathbf{x}$, we define the fitted conditional quantiles

$$q_j(\mathbf{x}) = \mathbf{x}^\top \hat{\theta}(\tau_j), \quad j = n-k, \ldots, m,$$

which serve as estimated upper order statistics of $F_Y(\cdot \mid \mathbf{x})$ and form the input to the tail index estimation in Section 2.1.4. Under standard quantile regression conditions (Wang et al., 2012, conditions A1–A2), these estimates are consistent and asymptotically normal. The tuning parameter k is discussed in Section 2.1.4.

In practice, estimated conditional quantiles may fail to be monotonically increasing in τ due to data sparsity or estimation error in the tail region. To enforce monotonicity, we apply the rearrangement procedure of Chernozhukov et al. (2010) to the sequence $\{q_j(\mathbf{x}),\ j = n-k, \ldots, m\}$ prior to tail index estimation.

For heavy-tailed distributions in the Fréchet domain with $\gamma(\mathbf{x}) > 0$, the tail of $F_Y(\cdot \mid \mathbf{x})$ decays at a polynomial rate governed by $\gamma(\mathbf{x})$. Inverting the tail approximation yields that for $\tau \to 1$, $Q_Y(\tau \mid \mathbf{x}) \approx c(\mathbf{x}) \cdot (1-\tau)^{-\gamma(\mathbf{x})}$, where $c(\mathbf{x})$ is a slowly varying function of $\mathbf{x}$. For this to be linear in $\mathbf{x}$, the exponent $-\gamma(\mathbf{x})$ must be the same for all $\mathbf{x}$, otherwise different covariate values produce different polynomial decay rates, which no linear function can capture. Wang and Li (2013, Proposition 2.1) formalise this as a necessary condition: linearity of $Q_Y(\tau \mid \mathbf{x})$ in $\mathbf{x}$ requires $\gamma(\mathbf{x})$ to be constant across $\mathbf{x}$, a property known as tail equivalence. When $\gamma(\mathbf{x})$ genuinely varies with $\mathbf{x}$, the linearity assumption in (2) is violated, motivating the power transformation introduced in Section 2.1.3.

2.1.3 Box-Cox Power Transformation

While linearity fails on the original scale when $\gamma(\mathbf{x})$ varies with $\mathbf{x}$, Wang and Li (2013, Example 1) show that for certain covariate-dependent EVI structures, linearity can be restored by an appropriate power transformation of Y. To restore linearity, we define the upper quantiles of the linear power-transformed QR model as

$$Q_{\Lambda_\lambda(Y)}(\tau \mid \mathbf{x}_i) = \mathbf{x}_i^\top \theta(\tau), \qquad \tau \in [1-\varepsilon, 1],$$

with the family of Box-Cox power transformations

$$\Lambda_\lambda(y) = \begin{cases} \dfrac{y^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log(y), & \lambda = 0. \end{cases}$$

The τth conditional quantile of y given $\mathbf{x}_i$ is then the inverse transformation $Q_Y(\tau \mid \mathbf{x}_i) = \Lambda_\lambda^{-1}\left(\mathbf{x}_i^\top \theta(\tau)\right)$.

For the estimation of the power transformation parameter λ, we adopt the method proposed in Mu and He (2007) and Yin et al. (2008), which estimates λ by

$$\hat{\lambda} = \underset{\lambda \in \mathbb{R}}{\arg\min} \sum_{i=1}^{n} \left\{ R_n(\mathbf{x}_i, \lambda; 1-\varepsilon) \right\}^2, \tag{3}$$

where $\tau = 1-\varepsilon$ with $\varepsilon > 0$ a small constant, and R_n a local lack-of-fit measure defined as

$$R_n(\mathbf{t}, \lambda; \tau) = \frac{1}{n} \sum_{j=1}^{n} I(\mathbf{x}_j \leq \mathbf{t}) \left[\tau - I\left\{ \Lambda_\lambda(y_j) - \mathbf{x}_j^\top \widehat{\theta}(\tau; \lambda) \leq 0 \right\} \right].$$

Here, $\mathbf{x} \leq \mathbf{t}$ denotes componentwise inequality, and $\widehat{\theta}(\tau; \lambda)$ is obtained by fitting linear quantile regression on the transformed responses at level τ:

$$\widehat{\theta}(\tau; \lambda) = \underset{\mathbf{b} \in \mathbb{R}^p}{\arg\min} \sum_{i=1}^{n} \rho_\tau\left(\Lambda_\lambda(y_i) - \mathbf{x}_i^\top \mathbf{b} \right),$$

where ρ_τ is the quantile loss function. The resulting estimator $\hat{\lambda}$ is root-n consistent under condition B5 of Wang and Li (2013).

2.1.4 Tail Index Estimation

Based on the MDA assumption (1), we estimate the covariate-dependent EVI $\gamma(\mathbf{x})$ using the Hill estimator (Hill, 1975), applied to the $k - [n^\eta]$ estimated upper order statistics $\{q_j(\mathbf{x})\}$:

$$\hat{\gamma}(\mathbf{x}) = \frac{1}{k - [n^\eta]} \sum_{j=[n^\eta]}^{k} \log\left(\frac{q_{n-j}(\mathbf{x})}{q_{n-k}(\mathbf{x})} \right),$$

where $q_j(\mathbf{x})$ are the fitted conditional quantiles at intermediate levels $\tau_j = j/(n+1)$, $j = n-k, \ldots, m$, as defined in Section 2.1.2. The sum is truncated at $j = [n^\eta]$, discarding quantile levels closest to 1 where fitted quantiles become unreliable due to data sparsity and where the Bahadur representation of $\hat{\theta}(\tau_j)$ may be inaccurate, breaking the asymptotic normality of $\hat{\gamma}$ (Wang et al., 2012).

The Hill estimator $\hat{\gamma}(\mathbf{x})$ is consistent and asymptotically normal (Wang et al., 2012, Theorem 3), and under the power transformation and an additional condition on the range of γ_0^*, this result also applies to the 3Stage estimator (Wang and Li, 2013, Theorem 3.1).

The parameter k represents the number of upper order statistics used in the Hill estimator, governing the trade-off between bias and variance. Smaller k yields higher variance as fewer observations inform the estimate, while larger k introduces bias by including observations outside the tail region. The asymptotic conditions $k \to \infty$ and $k/n \to 0$ ensure k grows with n while remaining a vanishing fraction of the sample. For the 2Stage estimator, Wang et al. (2012) use the fixed rule $k = [4.5 \cdot n^{1/3}]$. For the 3Stage estimator, Wang and Li (2013) select k by a data-driven criterion based on the identity $\gamma^*(\mathbf{x}) = \lambda \cdot \gamma(\mathbf{x})$, which can be estimated via two independent approaches. When k is chosen appropriately, the two estimators should agree, which motivates selecting k by minimising their integrated squared error:

$$k^* = \underset{k \geq 1}{\operatorname{argmin}} \sum_{i=1}^{n} \left[\widehat{\lambda\gamma}(\mathbf{x}_i) - \hat{\gamma}^*(\mathbf{x}_i) \right]^2. \tag{4}$$

The explicit forms of $\widehat{\lambda\gamma}(\mathbf{x})$ and $\hat{\gamma}^*(\mathbf{x})$ are given in Appendix A.

2.1.5 Extrapolation to Extreme Quantiles

To obtain the final extreme conditional quantile estimate of Y given $\mathbf{x}$ at target level $\tau_n \to 1$, we extrapolate from the intermediate quantile estimate $q_{n-k}(\mathbf{x})$ at level $\tau_{n-k} = \frac{n-k}{n+1}$ using the Weissman estimator (Weissman, 1978), adapted to the conditional case (Wang et al., 2012)

$$\hat{Q}_Y(\tau_n \mid \mathbf{x}) = \left(\frac{1-\tau_{n-k}}{1-\tau_n} \right)^{\hat{\gamma}(\mathbf{x})} q_{n-k}(\mathbf{x}).$$

This extrapolation relies on the Pareto tail approximation implied by the MDA assumption (1), where the quantile ratios follow a power law governed by $\hat{\gamma}(\mathbf{x})$.

Provided $np_n = o(k)$ and $\log(np_n) = o(k^{1/2})$, where $p_n = 1 - \tau_n$, the final estimator $\hat{Q}_Y(\tau_n \mid \mathbf{x})$ is asymptotically normal with convergence rate $\sqrt{k}/\log(k/(np_n))$ (Wang and Li, 2013, Theorem 3.2).

2.1.6 Overview of the Estimation Pipeline

Both estimators share the same general pipeline of intermediate conditional quantile estimation via NCS quantile regression, tail index estimation via the Hill estimator, and Weissman extrapolation to the target level τ_n.

The 2Stage estimator (Wang et al., 2012) applies this pipeline directly to Y, using $k = [4.5 \cdot n^{1/3}]$ as the number of upper order statistics.

The 3Stage estimator (Wang and Li, 2013) extends this pipeline by first estimating $\hat{\lambda}$ via the lack-of-fit criterion (3), fitting intermediate quantile regression on $\Lambda_{\hat{\lambda}}(Y)$, and back-transforming the fitted quantiles before tail index estimation, with k selected by the data-driven criterion k^* (4).

2.2 Empirical Evaluation Framework

2.2.1 Rolling Window Backtest Design

In our rolling window backtest, the object of interest is the conditional quantile estimate $\hat{Q}_Y(\tau|\mathbf{x}_t)$, which serves as the standard financial one-day Value-at-Risk (VaR) metric at confidence level τ (McNeil et al., 2005). A violation of the VaR occurs when the true return $r_t < -\hat{Q}_Y(\tau|\mathbf{x}_t)$.

The out-of-sample VaR forecasts are generated using a rolling estimation window of fixed length w. Let $\{r_t\}_{t=1}^N$ denote the full return series of length N. For a fixed window length $w < N$, the out-of-sample evaluation period runs from $t = w+1$ to $t = N$, which yields $T = N - w$ forecasts. At each evaluation date t, the one-step-ahead VaR forecast $\hat{Q}_t(\tau)$ is estimated strictly on the estimation window $\mathscr{W}_t = \{r_{t-w}, \ldots, r_{t-1}\}$. The window then shifts one step such that $\mathscr{W}_{t+1} = \{r_{t-w+1}, \ldots, r_t\}$.

The window length w governs the tradeoff between bias and variance. Longer windows include more observations for estimation, reducing parameter uncertainty, but at the cost of including older data that may no longer reflect the current tail behaviour. Smaller windows are more adaptive to structural change but increase estimation variance. The choice of w is therefore a critical modelling decision in our EVT-based methods, and we motivate it in Section 4.2.

2.2.2 GARCH-EVT Benchmark

The GARCH(1,1)-EVT model of McNeil and Frey (2000) combines a parametric volatility filter with a semi-parametric tail estimator. The conditional variance evolves as

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2,$$

with parameters estimated by quasi-maximum likelihood. The standardised residuals $\hat{z}_t = r_t/\hat{\sigma}_t$ are approximately i.i.d. under correct specification, and EVT is applied to their tail via the Peaks-over-Threshold method, fitting a Generalised Pareto Distribution above the 90th percentile threshold.

We use GARCH-EVT as our benchmark for two reasons. It is the established standard for dynamic VaR estimation with heavy-tailed returns, and it conditions on past information through the volatility forecast, providing a meaningful comparison with the QR-based estimators. The key difference is that GARCH-EVT conditions through a parametric variance equation and estimates tail shape unconditionally on residuals, whereas the QR-based methods condition nonparametrically through the covariate structure. This allows the latter to incorporate external covariates not yet reflected in past squared returns.

2.2.3 Regime Classification

Since the motivation for introducing the power transformation in the 3Stage estimator is the presence of nonlinearity in the conditional quantile function (Section 2.1.2), we classify rolling windows into regimes based on their degree of nonlinearity. This allows us to isolate and measure the empirical performance advantage of the power transformation in periods of higher nonlinearity, where the linearity assumption of the 2Stage estimator is more severely violated. We do not apply the EVI constancy test of Wang and Li (2013), which addresses the related but distinct question of whether $\gamma(\mathbf{x})$ varies with covariates, as its finite-sample reliability at $n = 500$ is limited at extreme

quantile levels. The nonlinearity score instead serves as a data-driven evaluation tool that partitions the sample by the degree of slope variation across quantile levels, without requiring a formal test.

We assign a nonlinearity score S_t to each forecast date t using the same estimation windows $\mathscr{W}_t$ defined in Section 2.2.1. The nonlinearity score S_t is computed as the total variation in covariate slopes, excluding the intercept, across quantile levels $\tau \in \{0.90, 0.92, 0.94, 0.96, 0.98, 0.99\}$,

$$S_t = \sum_{j=1}^{p} \sum_{l=1}^{L-1} \left| \hat{\beta}_j(\tau_{l+1}) - \hat{\beta}_j(\tau_l) \right|,$$

where $\hat{\beta}_j(\tau_l)$ is the estimated slope for covariate j at quantile level τ_l, $L = 6$ is the number of quantile levels, and p is the number of slope covariates.

We classify each forecast date t into one of three regimes based on its nonlinearity score: linear if the score falls below the 33rd percentile, middle if it lies between the 33rd and 67th percentiles, and nonlinear if it exceeds the 67th percentile.

2.2.4 Covariate Selection Procedure

To select the covariates for our model, we start by constructing a candidate set based on existing literature. All subsets of size one, two and three are then evaluated by their out-of-sample quantile loss at $\tau \in \{0.99, 0.995\}$, using the 2Stage estimator and the rolling window backtest design from Section 2.2.1. We restrict to subsets of at most three covariates to limit estimation noise in the intermediate quantile regression step, where additional parameters directly increase variance in the Hill estimator and Weissman extrapolation. The subset achieving the lowest average quantile loss across both quantile levels is selected as the final covariate set for our model. We use the 2Stage estimator at this stage since the intermediate quantile regression step is identical across both 2Stage and 3Stage, which makes it an exact substitute for covariate screening at a lower computational cost.

2.2.5 Statistical Evaluation Criteria

We use three criteria to evaluate estimator performance: the Kupiec unconditional coverage test as a calibration check, the average quantile loss for ranking estimators, and the Diebold–Mariano test for assessing the statistical significance of loss differences.

First, the Kupiec unconditional coverage test (Kupiec, 1995) uses a likelihood ratio statistic to test whether the observed violation rate is consistent with the target level τ, under the null hypothesis that violations occur independently with probability $1 - \tau$. We report the Kupiec test only for the full sample, where we use it exclusively as a calibration check, since at extreme quantile levels, the expected number of violations is small, limiting the test's power on separate regime subsamples. We therefore use it only as a sanity check that violation rates are broadly consistent with the target level, and rely on quantile loss for ranking the performance of the estimators.

Second, the quantile loss (Koenker and Bassett, 1978) is a strictly consistent scoring rule for the τ-th quantile, meaning that its expectation is minimised at the true conditional quantile. We therefore

rank estimators by the average quantile loss

$$\overline{\mathscr{L}}_\tau = \frac{1}{T}\sum_{t=1}^{T} \rho_\tau\left(r_t - \hat{Q}_t(\tau)\right),$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\})$. Unlike the Kupiec test, every observation contributes to $\overline{\mathscr{L}}_\tau$ regardless of whether a violation occurs, keeping the criterion informative at extreme quantile levels where violations are sparse.

Third, the Diebold–Mariano test (Diebold and Mariano, 1995) evaluates whether the observed difference in average quantile loss between two models is statistically significant, under the null hypothesis of equal predictive accuracy. The test statistic is asymptotically standard normal, and we use a HAC variance estimate (Newey and West, 1987) with lag truncation $\ell = w - 1$ to account for autocorrelation in the loss differentials.

3 Simulation Study

This section evaluates the finite-sample performance of the 2Stage and 3Stage estimators under a range of heavy-tailed data-generating processes. Section 3.1 introduces seven cases with diverse properties, and compares the estimators against direct quantile regression. Section 3.2 isolates the magnitude of EVI variation as a driver of 2Stage's failure through a targeted sensitivity analysis.

3.1 Simulation Cases

We consider seven simulation cases designed to stress test the 2Stage and 3Stage estimators across model misspecification and EVI heterogeneity. For Cases 1–5, the data are generated from a location-scale model on the Box-Cox transformed scale, with $\mathbf{x}_i = (1, x_{i1}, x_{i2})^\top$, $x_{ij} \sim$ Uniform$(-1,1)$ for $j = 1,2$, and $\beta = (2,1,1)^\top$ throughout. The inverse Box-Cox transformation Λ_λ^{-1}, defined in Section 2.1.3, maps each case back to the original scale.

Cases 6–7 are generated directly on the original scale from a Fréchet distribution with covariate-dependent EVI and a scalar covariate $x_i \sim$ Uniform$(-1,1)$. For these cases, the power-transformation model is misspecified for both the 2Stage and 3Stage estimators, since their conditional quantile functions cannot be linearised by any Box-Cox transformation. The case specifications are as follows.

- *Case 1*: Correctly specified, constant EVI.
  $\lambda = 1$, $\sigma = (1, 0.5, 0)^\top$, $\gamma(\mathbf{x}) = 1/2$, $\gamma^* = 1/2$, $\varepsilon_i \sim$ Pareto(2),
  $Q_Y(\tau \mid \mathbf{x}) = \Lambda_1^{-1}\left(\mathbf{x}^\top\beta + \mathbf{x}^\top\sigma \cdot (1-\tau)^{-1/2}\right)$.
- *Case 2*: Correctly specified after power transformation, constant EVI.
  $\lambda = 1/2$, $\sigma = (1, 0.5, 0)^\top$, $\gamma(\mathbf{x}) = 1/2$, $\gamma^* = 1/4$, $\varepsilon_i \sim$ Pareto(4),
  $Q_Y(\tau \mid \mathbf{x}) = \Lambda_{1/2}^{-1}\left(\mathbf{x}^\top\beta + \mathbf{x}^\top\sigma \cdot (1-\tau)^{-1/4}\right)$.
- *Case 3*: Correctly specified after log transformation, constant EVI.
  $\lambda = 0$, $\sigma = (1, 0.5, 0)^\top$, $\gamma(\mathbf{x}) = 1/2$, $\gamma^* = 0$, ε_i iid with $Q_\varepsilon(\tau) = \tau - 1 - \log(1-\tau)$,
  $Q_Y(\tau \mid \mathbf{x}) = \Lambda_0^{-1}\left(\mathbf{x}^\top\beta + \mathbf{x}^\top\sigma + \frac{1}{2}Q_\varepsilon(\tau)\right)$.

- *Case 4*: Correctly specified after power transformation, varying EVI.
  $\lambda = 1/2$, $\sigma = (0.5, 0.25, 0)^\top$, $\gamma(\mathbf{x}) = \mathbf{x}^\top\sigma \in [0.25, 0.75]$, $\gamma^*(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top\sigma \in [0.125, 0.375]$,
  $\varepsilon_i \sim \text{Pareto}(1/\gamma^*(\mathbf{x}_i))$,
  $Q_Y(\tau \mid \mathbf{x}) = \Lambda_{1/2}^{-1}\big(\mathbf{x}^\top\beta + (1-\tau)^{-\gamma(\mathbf{x})}\big)$.
- *Case 5*: Correctly specified after log transformation, varying EVI.
  $\lambda = 0$, $\sigma = (0.5, 0.25, 0)^\top$, $\gamma(\mathbf{x}) = \mathbf{x}^\top\sigma \in [0.25, 0.75]$, $\gamma^* = 0$, ε_i iid with
  $Q_\varepsilon(\tau) = \tau - 1 - \log(1-\tau)$,
  $Q_Y(\tau \mid \mathbf{x}) = \Lambda_0^{-1}\big(\mathbf{x}^\top\beta + \mathbf{x}^\top\sigma \cdot Q_\varepsilon(\tau)\big)$.
- *Case 6*: Misspecified, monotone varying EVI.
  $\gamma(x_i) = \exp(x_i - 1) \in [0.14, 1]$,
  $F_Y(y \mid x_i) = \exp\{-y^{-1/\gamma(x_i)}\}$, $Q_Y(\tau \mid x_i) = (-\log\tau)^{-\gamma(x_i)}$.
- *Case 7*: Misspecified, nonmonotone varying EVI.
  $\gamma(x_i) = 0.5 + 0.3\sin(\pi x_i) \in [0.2, 0.8]$,
  $F_Y(y \mid x_i) = \exp\{-y^{-1/\gamma(x_i)}\}$, $Q_Y(\tau \mid x_i) = (-\log\tau)^{-\gamma(x_i)}$.

We use the method from Section 2.1.3 to estimate λ for Cases 1–5, and additionally run the simulation given the true transformation parameter to isolate extrapolation error from λ estimation error. We estimate on quantile levels $\tau_n \in \{0.99, 0.995, 0.999\}$, use sample sizes $n \in \{500, 2000\}$, and repeat the simulation 500 times. To compare the performance of the estimators, we compute three metrics as Monte Carlo averages of the following per-replication quantities:

$$\text{IBias} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{Q}_Y(\tau_n \mid \mathbf{x}_i) - Q_Y(\tau_n \mid \mathbf{x}_i)\right],$$

$$\text{RIMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left[\hat{Q}_Y(\tau_n \mid \mathbf{x}_i) - Q_Y(\tau_n \mid \mathbf{x}_i)\right]^2},$$

$$\text{RMSPE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left[\frac{\hat{Q}_Y(\tau_n \mid \mathbf{x}_i) - Q_Y(\tau_n \mid \mathbf{x}_i)}{Q_Y(\tau_n \mid \mathbf{x}_i)}\right]^2}.$$

IBias captures the direction of systematic estimation error, RIMSE measures total estimation error combining both bias and variance on the original quantile scale, and RMSPE measures estimation error relative to the true quantile scale. Since the true quantile $Q_Y(\tau_n \mid \mathbf{x}_i)$ varies substantially in scale across cases, and across covariate values in cases where the EVI is non-constant, raw RIMSE is not comparable across cases or within Cases 4–7. The RMSPE (Hyndman and Koehler, 2006) addresses both problems. Though they do note that percentage-error measures are undefined when $Y_t = 0$, our DGPs have strictly positive support, so RMSPE is well-defined and finite for all cases. Therefore, we use the RMSPE as our primary metric and discuss IBias in the text where the direction of estimation error is meaningful.

We include the proposed 2Stage and 3Stage estimators, and for comparison, the QR estimator, which applies linear quantile regression directly on the original scale without tail extrapolation. For the 2Stage estimator we let $k = [4.5 \cdot n^{1/3}]$, and $\eta = 0.1$, as mentioned in Section 2.1.4 and Section 2.1.2, respectively. For the 3Stage estimator, we also let $\eta = 0.1$. For $n = 500$, we set $\varepsilon = 0.15$ and estimate k from the grid $[20, 70]$. For $n = 2000$, we set $\varepsilon = 0.1$ and estimate k from the grid $[30, 100]$.

Table 1 reports RMSPE across all seven cases, and the RIMSE and IBias tables are reported in Tables B1 and B2 in the Appendix. The results show that several patterns emerge consistently. The 3Stage estimator generally achieves the lowest RMSPE, and 2Stage is competitive with 3Stage under correct specification (Case 1), both dominating direct quantile regression (QR). As τ_n increases, all estimators degrade in accuracy, and the performance gaps widen, making the relative performance ordering clearer. The larger sample size of $n = 2000$ reduces RMSPE substantially across all methods and cases, and the 3Stage estimator recovers λ accurately at this sample size, confirming that instability observed at $n = 500$ and $\tau_n = 0.999$ reflects finite-sample noise rather than a structural feature of the estimators.

Cases 1, 2, and 3 share the same linear location-scale structure and constant EVI, differing only in the transformation parameter λ. The 2Stage estimator is mildly misspecified in Case 2 due to the nonlinear Box-Cox transformation at $\lambda = 1/2$, yet it performs comparably to 3Stage. Under Case 3, where $\lambda = 0$ implies an exponential relationship between the linear predictor and the conditional quantile on the original scale, 2Stage fails severely. This indicates that the linear intermediate quantile regression step in 2Stage can tolerate mild polynomial nonlinearity, but breaks down when the implied nonlinearity is exponential. Table B2 confirms that this failure corresponds to a large negative bias, reflecting systematic underestimation of the true conditional quantile at the intermediate regression step. In Cases 4 and 5, where $\gamma(\mathbf{x}) \in [0.25, 0.75]$ varies monotonically with covariates, and Case 6, where $\gamma(\mathbf{x}) \in [0.14, 1]$, the 2Stage estimator again breaks down. This is exactly where 3Stage outperforms both 2Stage and direct QR. In Case 7, where $\gamma(\mathbf{x}) = 0.5 + 0.3\sin(\pi x) \in [0.2, 0.8]$ varies nonmonotonically, 2Stage performs comparably to 3Stage even with a similar range of variation, pointing to the structure of EVI variation as a relevant factor.

Table 1: Root Mean Squared Percentage Error (RMSPE) of estimators of $Q_Y(\tau_n \mid \mathbf{x})$ across all cases, for $\tau_n \in \{0.99, 0.995, 0.999\}$ and $n \in \{500, 2000\}$.

	$n = 500$			$n = 2000$		
Method	$\tau_n = 0.99$	0.995	0.999	0.99	0.995	0.999
Case 1: $\lambda = 1$, constant EVI						
QR	0.3546	0.6558	1.2133	0.1550	0.2493	0.8468
2Stage	0.2129	0.2854	0.4945	0.1210	0.1765	0.3301
3Stage	0.2143	0.3121	2.8317	0.1231	0.1756	0.3360
3Stage (true λ)	0.2072	0.2788	0.4741	0.1213	0.1734	0.3262
Case 2: $\lambda = 1/2$, constant EVI						
QR	0.1971	0.3450	0.7337	0.1049	0.1589	0.5927
2Stage	0.1295	0.1668	0.2828	0.0893	0.1141	0.2149
3Stage	0.1245	0.1696	0.2978	0.0742	0.1104	0.2303
3Stage (true λ)	0.1224	0.1676	0.2957	0.0734	0.1101	0.2306
Case 3: $\lambda = 0$, constant EVI						
QR	1.7656	2.3974	3.0716	1.3119	1.4766	2.9013
2Stage	2.4×10^4	1.1×10^7	1.6×10^{13}	9.6285	3.3×10^3	4.4×10^9
3Stage	0.3295	0.4323	0.8162	0.1719	0.2280	0.3776
3Stage (true λ)	0.2960	0.3846	0.6676	0.1678	0.2260	0.3778
Case 4: $\lambda = 1/2$, varying EVI						
QR	4.1590	24.792	233.33	0.7690	1.6370	260.26
2Stage	10.919	205.98	1.9×10^5	1.3510	148.13	1.1×10^7
3Stage	0.7440	3.1501	216.90	0.3120	0.4580	1.7180
3Stage (true λ)	0.5540	0.8450	3.2080	0.3560	0.4980	1.0850
Case 5: $\lambda = 0$, varying EVI						
QR	3.6890	5.7220	17.862	2.5320	3.6200	19.223
2Stage	5.2×10^6	1.2×10^{10}	8.2×10^{17}	97.210	2.0×10^5	1.1×10^{13}
3Stage	0.4719	0.8450	4.9703	0.1775	0.2333	0.3880
3Stage (true λ)	0.2906	0.3751	0.6516	0.1655	0.2225	0.3690
Case 6: Fréchet, monotone varying EVI						
QR	2.0914	4.2433	26.053	1.3136	2.2585	13.490
2Stage	15.891	405.53	7.5×10^5	1.0875	1.9032	893.85
3Stage	0.2977	0.3607	0.5437	0.2102	0.2711	0.4675
Case 7: Fréchet, nonmonotone varying EVI						
QR	1.7397	3.9916	14.447	1.4383	2.2011	7.7614
2Stage	1.2106	1.6787	5.9311	1.2474	1.7003	3.3828
3Stage	0.9945	1.4126	4.4790	1.1090	1.6210	5.5295

3.2 Sensitivity Cases

To isolate the role of the magnitude of EVI variation, we replicate Case 4 with varying σ such that the range of $\gamma(\mathbf{x})$ widens across cases, while all other parameters remain identical. Specifically, $\lambda = 1/2$, $\beta = (2,1,1)^\top$, and $\gamma(\mathbf{x}) = \mathbf{x}^\top \sigma$ is centred at 0.5 in all three cases. The sensitivity cases are defined as:

- *Case S1:* $\sigma = (0.5, 0.05, 0)^\top$, such that $\gamma(\mathbf{x}) \in [0.45, 0.55]$ (near-constant EVI).
- *Case S2:* $\sigma = (0.5, 0.15, 0)^\top$, such that $\gamma(\mathbf{x}) \in [0.35, 0.65]$ (moderate EVI variation).
- *Case S3:* $\sigma = (0.5, 0.25, 0)^\top$, such that $\gamma(\mathbf{x}) \in [0.25, 0.75]$ (large EVI variation).

Table 2 reports RMSPE for Cases S1–S3, with companion RIMSE and IBias tables in Tables B3 and B4 in the Appendix. At $n = 500$, a clear threshold emerges. At $\tau_n = 0.995$, 2Stage RMSPE rises from 0.96 in Case S1 to 157.9 in Case S2 and 206.0 in Case S3, while 3Stage remains stable, and this ordering holds across all quantile levels. At $n = 2000$, the pattern is less uniform. In Case S2, 2Stage performs comparably to 3Stage at $\tau_n = 0.99$ but deteriorates sharply at more extreme levels, suggesting that larger samples partially mitigate 2Stage's sensitivity to moderate EVI variation at less extreme quantiles, while the failure at $\tau_n = 0.995$ and $\tau_n = 0.999$ persists. Taken together, 2Stage tolerates near-constant EVI variation but fails once the variation width reaches roughly 0.30, with the failure threshold shifting to smaller widths as τ_n increases.

Table 2: Root Mean Squared Percentage Error (RMSPE) of estimators of $Q_Y(\tau_n \mid \mathbf{x})$ across sensitivity cases S1–S3, for $\tau_n \in \{0.99, 0.995, 0.999\}$ and $n \in \{500, 2000\}$. All cases replicate Case 4 with $\lambda = 1/2$, $\beta = (2,1,1)^\top$, varying only σ.

	$n = 500$			$n = 2000$		
Method	$\tau_n = 0.99$	0.995	0.999	0.99	0.995	0.999
Case S1: $\gamma(\mathbf{x}) \in [0.45, 0.55]$, *near-constant EVI*						
QR	0.9188	3.3320	11.140	0.2947	0.5307	3.3290
2Stage	0.4421	0.9567	17.939	1.0960	82.262	1.8×10^6
3Stage	0.4036	0.5655	1.5961	0.2277	0.3250	0.7227
3Stage (true λ)	0.3759	0.4963	0.8837	0.2255	0.3150	0.5863
Case S2: $\gamma(\mathbf{x}) \in [0.35, 0.65]$, *moderate EVI variation*						
QR	1.0747	2.7668	59.377	0.3695	0.7174	31.421
2Stage	7.6306	157.89	1.7×10^5	0.3373	4.6210	1.7×10^4
3Stage	1.3250	1.5890	9.2581	0.2581	0.4413	15.858
3Stage (true λ)	0.4116	0.5754	1.2754	0.2405	0.3332	0.6105
Case S3: $\gamma(\mathbf{x}) \in [0.25, 0.75]$, *large EVI variation*						
QR	4.1590	24.792	233.33	0.7690	1.6370	260.26
2Stage	10.919	205.98	1.9×10^5	1.3510	148.13	1.1×10^7
3Stage	0.7440	3.1501	216.90	0.3120	0.4580	1.7180
3Stage (true λ)	0.5540	0.8450	3.2080	0.3560	0.4980	1.0850

4 Empirical Application: Bitcoin Tail Risk

In this section, we apply the estimators described above to daily Bitcoin return data to estimate extreme Value-at-Risk (VaR) forecasts. Our particular interest lies in evaluating whether the power-transformed 3Stage estimator of Wang and Li (2013) produces more accurate extreme VaR forecasts than the 2Stage estimator (Wang et al., 2012) and the GARCH(1,1)-EVT benchmark from Section 2.2.2, and additionally whether any performance advantages concentrate in periods of higher nonlinearity. To examine this, we use the empirical evaluation framework from Section 2.2.

4.1 Data

Daily Bitcoin (BTC) price data are obtained from Yahoo Finance (ticker: BTC-USD, via the `yfinance` Python package; `https://finance.yahoo.com`), covering 2015–2026 with roughly 4,170 observations, and displayed in Figure 1. We use 2015–2018 (roughly 1,460 observations) as the screening period for covariate selection, and 2019–2026 (roughly 2,710 observations) as the out-of-sample evaluation period. The log returns are then defined by $r_t = \log(P_t/P_{t-1})$, where P_t is the price of BTC at time t. For VaR estimation, we work with losses $\ell_t = -r_t$, so that a VaR is violated when $\ell_t > \hat{Q}_t(\tau)$. The Box-Cox transformation in the 3Stage estimator requires strictly positive inputs. We therefore apply, within each estimation window, a location shift $a_t = -\min_{s \in W_t}(\ell_s) + \delta$, with $\delta > 0$ a small constant ensuring strict positivity, and recover the VaR on the original loss scale by subtracting a_t from the returned estimate, which is exact by quantile equivariance (Koenker and Bassett, 1978).

The choice of covariates for the QR-based estimators is guided by existing literature on Bitcoin tail risk drivers. The extensive covariate selection of Kwon (2021) and Lawuobahsumo et al. (2024) concludes that in the case of 5%-VaR, volatility-related variables drive the most variation, consistent with GARCH-like filtering. However, at the 1%-VaR, macroeconomic variables tend to become dominant drivers, suggesting that extreme tail events are driven by systematic macroeconomic risk, and not just by local volatility clustering. Standard volatility models miss this channel. In practice, the framework of our 3Stage and 2Stage estimators requires limiting the number of covariates, as additional parameters directly increase variance in the Hill estimator and Weissman extrapolation steps.

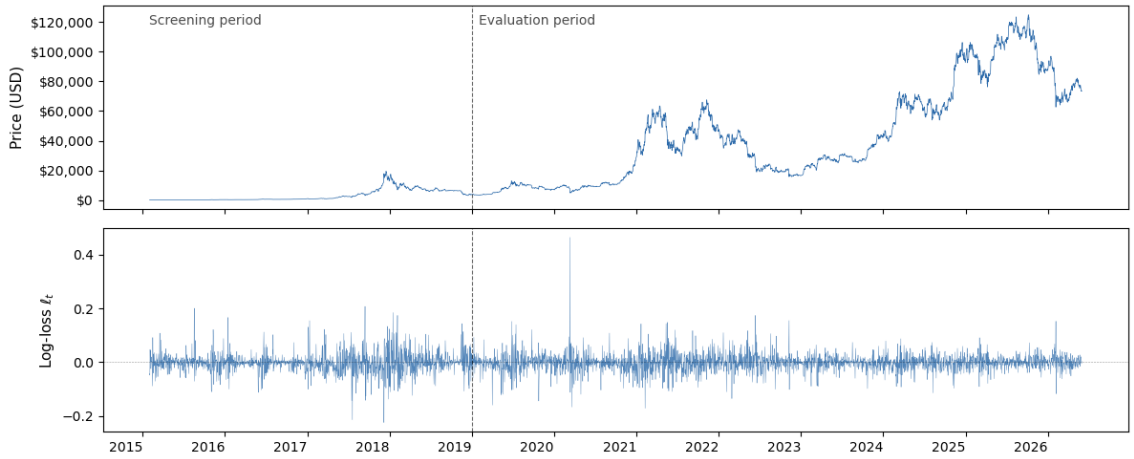


Figure 1: Daily Bitcoin (BTC-USD) price (top) and log-loss series (bottom).

Furthermore, since we are estimating daily forecasts, only daily-frequency covariates are eligible, ruling out some macroeconomic variables. Our candidate set, displayed in Table 3, therefore consists of the best-performing daily-frequency variables from the literature discussed above, along with two market-specific covariates. First, the 30-day realised volatility captures the conditional variance channel also captured by GARCH. Second, the maximum drawdown measures the magnitude of recent price corrections.

To select the set of covariates, we conduct the out-of-sample quantile loss test described in Section 2.2.4, which uses the strict screening period 2015–2018. The selected covariates are then fixed, and all models are evaluated solely on the period 2019–2026, ensuring strict separation between covariate selection and model evaluation. The selected covariates are: lagged log return (`lag_return`) and 5-year forward inflation expectation (`T5YIFR`). This subset achieved the lowest average quantile loss score across $\tau \in \{0.99, 0.995\}$ among all size 1, 2, and 3 subsets evaluated.

4.2 Window Length Selection

As mentioned in Section 2.2.1, the window length w governs the tradeoff between bias and variance, but since the Bitcoin data has non-stationary tail behaviour, the window length must also be responsive to Bitcoin's structural breaks in its volatility regimes. Larger windows risk blurring structural breaks by averaging over multiple volatility regimes, while smaller windows stay more responsive to regime changes.

We set our candidate lower bound at $w = 500$, since our simulation study in Section 3.1 shows reliable estimates for the 3Stage and 2Stage estimators at $n = 500$, and we cap our upper bound at $w = 2000$ as this is the largest sample size considered in our simulation study, and windows beyond this would span multiple volatility regimes. Our out-of-sample evaluation period has a total of $N = 2{,}710$ observations, which in practice yields $N = 2{,}708$ observations, since the log-differenced EPU covariate requires an additional lag and drops the first two evaluation dates. For $w = 500$, this gives $T = 2{,}208$ forecasts, and for $w = 2000$, we still have $T = 708$ forecasts, confirming that evaluation power is not a binding constraint for our primary specification.

We therefore balance between estimation reliability, which pushes w up, and regime responsiveness, which pushes w down. Chaim and Laurini (2018) identify two high-volatility regimes in Bitcoin over 2013–2018, with a calm period in between of roughly 500 trading days. Therefore, a window of $w = 500$ sits comfortably within a single regime during calmer periods, while remaining short enough to transition between regimes during turbulent ones. Although their sample ends in 2018 while ours extends to 2026, the additional regime shifts over 2018–2026 further motivate a responsive rather than a large window. We therefore set $w = 500$ as our primary specification.

4.3 Diagnostic Checks

To investigate whether the 3Stage and 2Stage estimators are fit for the Bitcoin data, we apply three diagnostic checks. First, we test whether Bitcoin losses are in the Fréchet MDA, that is, $\gamma > 0$, which is necessary since the entire 3Stage and 2Stage methodology rests on this assumption (Section 2.1.1). Second, we check the QQ plot to assess how well the Pareto approximation fits the actual tail. Finally, we investigate if the quantile slopes are nonlinear. Nonlinearity is exactly the motivation for proposing the 3Stage model as an improvement on the 2Stage estimator, and is

Table 3: Candidate covariate set for the empirical application.

Covariate	Symbol	Source	Economic motivation
Panel A: Market-specific covariates			
30-day realised volatility	$\sigma_{30,t-1}$	Yahoo Finance (BTC-USD)	Captures the conditional variance channel underlying GARCH-type models. High realised volatility signals volatility clustering and elevated tail risk.
Maximum drawdown from all-time high	DD_{t-1}	Yahoo Finance (BTC-USD)	Measures the cumulative magnitude of price corrections. Deep drawdowns signal distressed market conditions and elevated crash risk (Chaim and Laurini, 2018).
Lagged log-return	r_{t-1}	Yahoo Finance (BTC-USD)	Controls for short-horizon momentum and mean-reversion dynamics in cryptocurrency returns (Kwon, 2021).
Panel B: Macroeconomic and financial covariates			
CBOE Volatility Index	VIX_{t-1}	FRED (`VIXCLS`)	Measures implied volatility in equity markets. Increased market stress drives capital reallocation away from speculative assets, amplifying Bitcoin tail risk (Lawuobahsumo et al., 2024).
Treasury yield spread (10Y−3M)	TS_{t-1}	FRED (`T10Y3M`)	An inverted yield curve signals deteriorating macroeconomic conditions and broader financial-market stress (Lawuobahsumo et al., 2024).
5-year forward inflation expectation	π^{5y}_{t-1}	FRED (`T5YIFR`)	Forward inflation expectations capture monetary-policy uncertainty. Rapid regime shifts are linked to increased tail risk across risk assets (Lawuobahsumo et al., 2024).
EUR/USD exchange rate	FX_{t-1}	FRED (`DEXUSEU`)	Broad dollar-strength indicator. Dollar appreciation is associated with global risk-off episodes and cryptocurrency sell-offs (Kwon, 2021).
Economic Policy Uncertainty index (log-diff.)	$\Delta\mathrm{EPU}_{t-1}$	FRED (`USEPUINDXD`)	Day-over-day change in log-EPU captures sudden increases in policy uncertainty. Found to be a significant predictor of crypto tail risk at the 1% level (Kwon, 2021).
Investment-grade corporate bond return	CB_{t-1}	Yahoo Finance (LQD ETF)	Daily return on the iShares iBoxx IG Corporate Bond ETF. Captures credit-spread dynamics and risk sentiment changes in fixed-income markets (Kwon, 2021; Lawuobahsumo et al., 2024).

Notes: All variables are lagged by one trading day to ensure strict predeterminedness with respect to the day-t log-loss, and availability in the information set $\mathscr{F}_{t-1}$ when forecasting. FRED-sourced covariates are not observed on weekends or holidays. On these days, the most recently available business-day value is carried forward before applying the one-day lag. Yahoo Finance data retrieved via the `yfinance` Python package (`https://finance.yahoo.com`); FRED data retrieved via the Federal Reserve Bank of St. Louis (`https://fred.stlouisfed.org`). The final specification is selected by the out-of-sample quantile loss screen described in Section 2.2.4.

therefore required for a meaningful comparison in our empirical application.

To check if Bitcoin losses are in the Fréchet MDA, we plot the Hill estimator (Section 2.1.4) $\hat{\gamma}(k)$ against k for the full unconditional loss series $\ell_1, \ldots \ell_N$, and look for a stable plateau at a positive value. Here, k is the number of upper order statistics used in the Hill estimator. Figure 2 reveals $\hat{\gamma}(k)$ stabilising around 0.37, confirming that $\gamma > 0$ (Fréchet MDA), and that the estimate is not excessively sensitive to the choice of k, confirming that a reliable estimate exists. We operate this diagnostic check on the unconditional loss distribution because applying it directly on the conditional tail is not feasible since this is the object we are trying to estimate. We verify the unconditional tail and argue that the conditional tail inherits this property given that the covariates are bounded (McNeil and Frey, 2000). The stability of the produced rolling $\hat{\gamma}(\mathbf{x}_t)$ estimates during the backtest serves as a conditional confirmation.

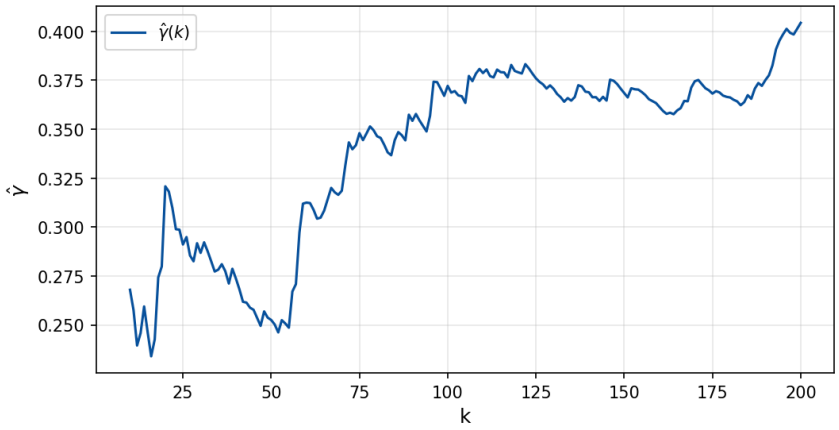


Figure 2: Hill plot of BTC daily losses.

We concluded from our Hill plot that $\gamma > 0$, but to investigate how well the Pareto distribution actually fits the shape of the tail across quantiles, we compute the QQ plot, displayed in Figure 3. The figure plots the empirical tail quantiles of the losses $\ell_1, \ldots \ell_N$ against the theoretical quantiles of a Pareto distribution. The tail follows a power law closely across most quantiles, but is somewhat lighter than the fitted Pareto at very extreme values. This confirms that the tail of $\ell_1, \ldots \ell_N$ is consistent with a Pareto distribution, supporting the Fréchet MDA assumption. The slight undershoot at the very extremes suggests that EVT extrapolation at $\tau = 0.999$ may be conservative, which is preferable to underestimating tail risk.

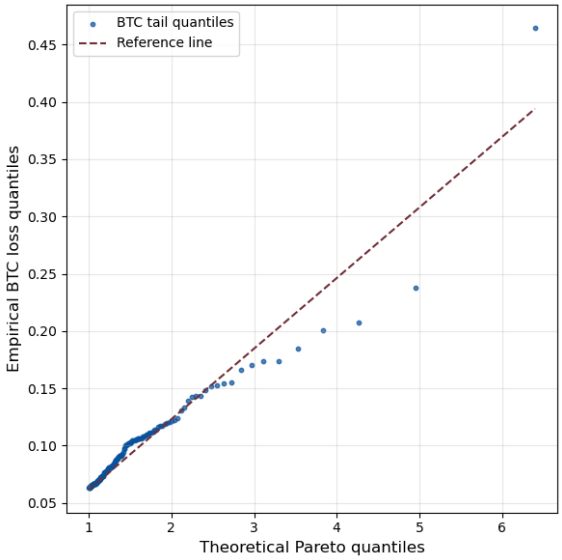


Figure 3: Pareto QQ plot of BTC daily losses.

To investigate nonlinearity in the quantile slopes, we plot the estimated slope $\hat{\beta}(\tau)$ of the 30-day realised volatility and the drawdown across $\tau \in [0.90, 0.99]$. Figure 4 shows substantial nonlinear variation in the slopes, such that the linearity assumption of Wang et al. (2012) is violated, motivating the power transformation in Wang and Li (2013). The remaining covariates exhibit similar behaviour.

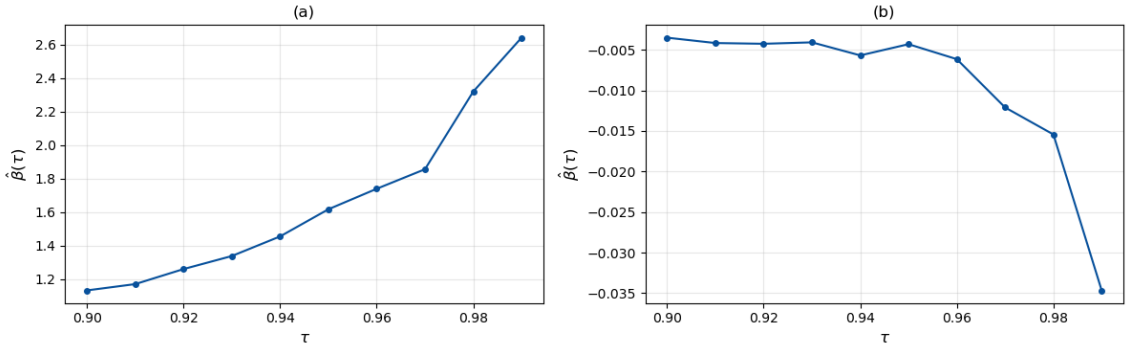


Figure 4: Estimated quantile slopes $\hat{\beta}(\tau)$ for 30-day realised volatility (a) and the drawdown (b) across $\tau \in [0.90, 0.99]$.

4.4 Empirical Application Results

Table 4 reports full-sample violation rates, Kupiec p-values, and average quantile loss for all three estimators. The Kupiec unconditional coverage test is used solely as a calibration check, as noted in Section 2.2.5. At $\tau = 0.99$, we expect roughly 22 violations, and the test has reasonable power. At this level, no estimator rejects the null of correct unconditional coverage. At $\tau = 0.995$, we expect roughly 11 violations, and the test has limited but non-negligible power. Only the 3Stage estimator falls just short of the 5% significance threshold ($p = 0.053$), while GARCH-EVT and 2Stage pass comfortably ($p = 0.990$). Finally, at the highest level of $\tau = 0.999$, only roughly 2 violations are expected, making the test nearly uninformative. The 3Stage rejection at this level ($p = 0.003$)

Table 4: Full-sample backtest results at $w = 500$. Violation rate is the fraction of out-of-sample forecasts for which the realised return falls below $-\hat{Q}_t(\tau)$. Kupiec p is the p-value of the unconditional coverage test (Kupiec, 1995). QL denotes average quantile loss.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	GARCH-EVT	2Stage	3Stage
$\tau = 0.990$			
Viol. rate	1.13%	1.18%	1.36%
Kupiec p	0.541	0.415	0.104
QL	0.001085	0.001135	0.001157
$\tau = 0.995$			
Viol. rate	0.50%	0.50%	0.82%
Kupiec p	0.990	0.990	0.053*
QL	0.000629	0.000669	0.000702
$\tau = 0.999$			
Viol. rate	0.18%	0.14%	0.36%
Kupiec p	0.279	0.613	0.003***
QL	0.000193	0.000214	0.000222

reflects an excess of observed violations consistent with the finite-sample instability documented in the simulation study at $n = 500$ (Section 3.1), and not a result of systematic miscalibration. Overall, the Kupiec tests, at levels at which they have power, do not point to problematic calibration.

GARCH-EVT produces the lowest average QL across quantile levels, followed by 2Stage, with 3Stage scoring the highest. The differences are, however, small. For example, at $\tau = 0.995$, the gap between GARCH-EVT and 3Stage is 0.000073, and the Diebold-Mariano tests reveal no significant differences in predictive accuracy between any pair of estimators.

We conclude that on the full sample, the results do not distinguish among the estimators, and we therefore turn to the regime-conditional results in Tables 5 and 6. The findings show a clear regime-dependent pattern in GARCH-EVT's performance advantage. In the middle regime, GARCH-EVT significantly outperforms both covariate-conditional estimators at $\tau = 0.99$, and significantly outperforms 3Stage at $\tau = 0.995$. In the linear regime, GARCH-EVT's advantage is directionally consistent but does not reach significance at any τ level, and in the nonlinear regime, GARCH-EVT also does not perform significantly better than either covariate-conditional estimator at any quantile level. GARCH-EVT's performance edge is therefore concentrated in the middle regime, where volatility clustering appears most informative and where GARCH-EVT holds a significant advantage over 2Stage or 3Stage in two of the nine regime–quantile combinations, while in the linear and nonlinear regimes, the advantage does not reach significance at any level.

Table 5: Regime-conditional average quantile loss (QL) at $\tau \in \{0.99, 0.995, 0.999\}$, $w = 500$. Regimes are defined by a tertile split of the nonlinearity score S_t (Section 2.2.3). Pairwise Diebold–Mariano tests are reported in Table 6.

Regime	GARCH-EVT	2Stage	3Stage
$\tau = 0.990$			
Linear ($T = 728$)	0.001071	0.001103	0.001154
Middle ($T = 751$)	0.001133	0.001240	0.001276
Nonlinear ($T = 729$)	0.001051	0.001058	0.001037
$\tau = 0.995$			
Linear ($T = 728$)	0.000574	0.000628	0.000673
Middle ($T = 751$)	0.000693	0.000763	0.000825
Nonlinear ($T = 729$)	0.000618	0.000613	0.000604
$\tau = 0.999$			
Linear ($T = 728$)	0.000162	0.000217	0.000217
Middle ($T = 751$)	0.000244	0.000243	0.000278
Nonlinear ($T = 729$)	0.000171	0.000181	0.000167

Table 6: Diebold–Mariano tests (Diebold and Mariano, 1995) for regime-conditional predictive accuracy (HAC, $\ell = 499$). A negative DM statistic indicates the first-named model has lower quantile loss. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	GARCH vs 2Stage	3Stage vs 2Stage	3Stage vs GARCH
$\tau = 0.990$			
Linear ($T = 728$)	-1.450 (0.147)	$+1.049$ (0.294)	$+1.548$ (0.122)
Middle ($T = 751$)	-5.698 ($< 0.001^{***}$)	$+2.077$ (0.038^{**})	$+4.558$ ($< 0.001^{***}$)
Nonlinear ($T = 729$)	-0.195 (0.845)	-3.115 (0.002^{***})	-0.307 (0.759)
$\tau = 0.995$			
Linear ($T = 728$)	-1.517 (0.129)	$+0.945$ (0.345)	$+1.304$ (0.192)
Middle ($T = 751$)	-1.570 (0.116)	$+2.184$ (0.029^{**})	$+4.246$ ($< 0.001^{***}$)
Nonlinear ($T = 729$)	$+0.412$ (0.680)	-0.619 (0.536)	-0.989 (0.322)
$\tau = 0.999$			
Linear ($T = 728$)	-1.058 (0.290)	$+0.063$ (0.950)	$+1.098$ (0.272)
Middle ($T = 751$)	$+0.034$ (0.973)	$+1.182$ (0.237)	$+0.463$ (0.644)
Nonlinear ($T = 729$)	-0.836 (0.403)	-4.194 ($< 0.001^{***}$)	-0.370 (0.711)

Moreover, the performance advantage of GARCH-EVT in the middle regime becomes less pronounced as τ increases: significant against both 2Stage and 3Stage at $\tau = 0.99$, significant against 3Stage only at $\tau = 0.995$, and absent entirely at $\tau = 0.999$. This indicates that the covariate-conditional estimators close the gap with GARCH-EVT at more extreme quantile levels, which is consistent with the evidence documented by Kwon (2021) and Lawuobahsumo et al. (2024), highlighted in Section 1. They show that macroeconomic variables become stronger drivers of Bitcoin tail risk at more extreme quantile levels, a process that volatility clustering can only partially explain. However, the power of the DM test is weak at $\tau = 0.999$ due to sparse observations, so this cannot be interpreted as definitive evidence of convergence.

The 3Stage estimator has an edge over 2Stage only in the nonlinear regime, with statistical significance at $\tau = 0.99$ and $\tau = 0.999$. In the middle regime, 2Stage significantly outperforms

3Stage at $\tau = 0.99$ and $\tau = 0.995$. While these differences are statistically significant, the absolute quantile loss differences remain small. At $\tau = 0.99$ in the nonlinear regime, 2Stage's quantile loss exceeds that of 3Stage by roughly 2%, which is not as dramatic as observed in the simulation study (Section 3.1). This points to the Bitcoin data not containing as much misspecification as the most extreme simulation cases. The modest performance gap may be explained by the nonlinearity of the conditional quantile function itself, which the simulation study shows 2Stage can tolerate when the implied nonlinearity is mild and polynomial. A different cause could be a covariate-varying EVI, where the sensitivity analysis (Section 3.2) shows that 2Stage fails severely once the degree of EVI variation exceeds a moderate threshold, but remains competitive under near-constant EVI. The performance gap in the empirical results is small enough to suggest that EVI variation in the Bitcoin data falls within the range that 2Stage can accommodate, though the two channels cannot be cleanly separated from the empirical results alone.

Overall, the regime-conditional results suggest a narrow but significant advantage for GARCH-EVT in the middle regime, while the covariate-conditional estimators are most competitive in the linear and nonlinear regimes and at more extreme quantile levels, with 3Stage outperforming 2Stage in areas where nonlinearity is more pronounced.

5 Conclusion

This paper assessed whether covariate-conditional EVT-QR (2Stage, 3Stage) can compete with GARCH-EVT for extreme VaR forecasting of Bitcoin using a rolling window backtest design. Research has shown that the extreme tails of Bitcoin are mainly driven by macroeconomic factors (Kwon, 2021; Lawuobahsumo et al., 2024), and the covariate-conditional estimators can incorporate this information directly, whereas GARCH-EVT captures it only partially through volatility clustering. At more extreme quantile levels, 2Stage and 3Stage may therefore hold an advantage that empirical results partially support.

In the simulation study, the main failure modes revealed for 2Stage are exponential nonlinearity in the conditional quantile function and covariate-varying EVI. While the method was robust to mild polynomial nonlinearity with constant EVI, it failed under exponential nonlinearity and worsened with larger variation in the EVI. The sensitivity analysis additionally showed that the EVI variation threshold at which 2Stage fails narrows as τ increases, so that a moderate amount of variation that was tolerable at $\tau = 0.99$ becomes problematic at $\tau = 0.995$ and $\tau = 0.999$. 3Stage resolves both failure modes, yet at sample sizes comparable to the empirical setting, it exhibits finite-sample instability that undermines its practical advantage.

In the empirical application to Bitcoin, the results showed no significant performance differences among estimators across the full sample, while the regime-conditional analysis showed GARCH-EVT to have a significant advantage in the middle regime. This advantage decreased with quantile levels, with no significant advantage remaining at $\tau = 0.999$, which could indicate that the covariate-conditional estimators close the gap with GARCH-EVT at more extreme quantile levels. However, since the Diebold-Mariano test has limited power at $\tau = 0.999$, investigation using larger samples would be needed to confirm whether the covariate-conditional estimators genuinely narrow the performance differential.

Turning to the comparison between EVT-QR methods, the findings demonstrate that 3Stage

significantly outperforms 2Stage only in the nonlinear regime at $\tau = 0.99$ and $\tau = 0.999$, while 2Stage outperforms 3Stage in the middle regime at $\tau = 0.99$ and $\tau = 0.995$. The modest absolute performance gap suggests that the Bitcoin data do not exhibit the degree of misspecification that drives 2Stage's severe failure in the most extreme simulation cases. Since the nonlinearity score defined in Section 2.2.3 cannot distinguish the source of misspecification, further investigation is needed to determine whether the modest gap stems from mild polynomial nonlinearity or limited EVI variation. Identifying the source matters for model selection, as both channels can intensify under different market conditions and push 2Stage toward the same breakdown observed in simulation.

The results suggest that covariate-conditional EVT-QR is competitive with GARCH-EVT in the linear and nonlinear regimes, and that 3Stage in particular offers an advantage over 2Stage under nonlinear conditions, but the framework does not offer a general improvement over GARCH-EVT at this sample size, possibly reflecting degrees-of-freedom constraints that limit the effective number of tail observations available for extreme quantile estimation. GARCH-EVT's advantage is instead concentrated in the middle regime. Model selection should therefore incorporate market regimes rather than applying a single model uniformly, and the nonlinearity score defined in Section 2.2.3 may serve as a practical indicator for switching between models. Using the score for this purpose, however, requires further validation beyond the scope of this thesis.

Several directions remain open, including extending the framework to the estimation of Expected Shortfall and testing generalisability across other assets with high volatility. Addressing 3Stage's finite-sample instability documented in Section 3.1 is also a natural next step. Further, since strongly varying EVI drives 2Stage's severe estimation breakdown, and the nonlinearity score alone cannot identify whether covariate-varying EVI is the source, improving the finite-sample reliability of the EVI constancy test proposed by Wang and Li (2013) would help identify settings where covariate-conditional estimators hold a meaningful advantage over existing models.

Appendix

A Data-Driven Selection of k

The data-driven criterion for k in the 3Stage estimator is based on the identity $\gamma^*(\mathbf{x}) = \lambda \cdot \gamma(\mathbf{x})$, which admits two independent estimates. The first scales the Hill estimator by $\hat{\lambda}$:

$$\widehat{\lambda\gamma}(\mathbf{x}) = \hat{\lambda} \cdot M_{0,n}^{(1)},$$

and the second uses the moment estimator of de Haan and Ferreira (2006):

$$\hat{\gamma}^*(\mathbf{x}) = M_{0,n}^{(1)} + 1 - \frac{1}{2}\left(1 - \frac{\left(M_{0,n}^{(1)}\right)^2}{M_{0,n}^{(2)}}\right)^{-1},$$

where

$$M_{0,n}^{(i)} = \frac{1}{k - [n^\eta]} \sum_{j=[n^\eta]}^{k} \left\{ \log \frac{\mathbf{x}^\top \hat{\theta}(\tau_{n-j}; \hat{\lambda})}{\mathbf{x}^\top \hat{\theta}(\tau_{n-k}; \hat{\lambda})} \right\}^i, \quad i = 1, 2.$$

B Simulation Tables

Table B1: Root Integrated Mean Squared Error (RIMSE) of estimators of $Q_Y(\tau_n \mid \mathbf{x})$ across all cases, for $\tau_n \in \{0.99, 0.995, 0.999\}$ and $n \in \{500, 2000\}$. Companion to Table 1

	$n = 500$			$n = 2000$		
Method	$\tau_n = 0.99$	0.995	0.999	0.99	0.995	0.999
Case 1: $\lambda = 1$, constant EVI						
QR	4.3449	10.555	39.903	1.9107	3.9624	27.337
2Stage	2.6771	4.7441	15.955	1.5557	3.0296	11.547
3Stage	2.7677	5.0184	52.093	1.5927	3.0484	11.818
3Stage (true λ)	2.6529	4.7439	16.002	1.5643	3.0116	11.645
Case 2: $\lambda = 1/2$, constant EVI						
QR	2.4189	4.6201	16.561	1.3563	2.3025	10.729
2Stage	1.9239	2.9978	8.0397	1.2911	2.0459	6.1697
3Stage	1.8110	2.9459	8.1623	1.0470	1.8554	6.1560
3Stage (true λ)	1.7761	2.9035	8.0763	1.0260	1.8422	6.1703
Case 3: $\lambda = 0$, constant EVI						
QR	306.59	561.98	1570.9	252.53	379.13	1527.2
2Stage	7.7×10^5	4.9×10^8	1.6×10^{15}	313.38	9.7×10^4	2.9×10^{11}
3Stage	276.24	540.03	2763.4	99.896	182.55	667.21
3Stage (true λ)	164.53	306.64	1269.1	94.502	178.78	668.78
Case 4: $\lambda = 1/2$, varying EVI						
QR	251.88	2320.1	6.1×10^4	56.642	166.88	3.6×10^4
2Stage	114.69	2215.9	3.9×10^6	58.751	1824.3	2.1×10^8
3Stage	80.807	259.10	7097.5	47.294	169.20	7597.4
3Stage (true λ)	62.811	183.08	2019.4	48.688	151.20	1825.0
Case 5: $\lambda = 0$, varying EVI						
QR	217.73	402.59	2060.5	180.16	312.61	2765.9
2Stage	2.3×10^7	6.6×10^{10}	6.8×10^{18}	456.24	1.0×10^6	8.4×10^{13}
3Stage	387.19	1331.2	2.9×10^4	78.959	162.92	865.93
3Stage (true λ)	110.65	230.92	1211.7	65.468	141.65	716.50
Case 6: Fréchet, monotone varying EVI						
QR	19.154	46.857	443.34	15.640	32.195	248.61
2Stage	34.124	834.78	1.9×10^6	15.935	32.661	2286.0
3Stage	8.6305	19.800	162.47	5.3278	13.822	144.85
Case 7: Fréchet, nonmonotone varying EVI						
QR	12.239	38.201	256.00	9.7887	18.735	116.89
2Stage	10.665	20.049	171.85	9.7973	18.250	73.820
3Stage	14.617	30.596	222.51	15.162	33.666	285.10

Table B2: Integrated Bias (IBias) of estimators of $Q_Y(\tau_n \mid \mathbf{x})$ across all cases, for $\tau_n \in \{0.99, 0.995, 0.999\}$ and $n \in \{500, 2000\}$. Companion to Table 1

	$n = 500$			$n = 2000$		
Method	$\tau_n = 0.99$	0.995	0.999	0.99	0.995	0.999
Case 1: $\lambda = 1$, constant EVI						
QR	0.7472	2.0172	−2.2923	0.1237	0.3225	4.5005
2Stage	−0.8454	−2.1291	−9.6661	−0.4913	−1.4704	−7.8563
3Stage	−0.9467	−2.2973	−9.9848	−0.4978	−1.5969	−8.5448
3Stage (true λ)	−0.9532	−2.3141	−10.163	−0.5048	−1.6050	−8.5737
Case 2: $\lambda = 1/2$, constant EVI						
QR	0.3610	0.6019	−2.3737	0.1365	0.2185	1.5739
2Stage	−0.5178	−1.2750	−5.2353	−0.1835	−0.7250	−3.9930
3Stage	−0.7261	−1.5486	−5.7030	−0.3016	−0.9004	−4.3428
3Stage (true λ)	−0.7065	−1.5144	−5.6220	−0.2981	−0.8992	−4.3491
Case 3: $\lambda = 0$, constant EVI						
QR	83.547	128.69	80.733	54.230	83.741	294.82
2Stage	−1950.6	-1.1×10^6	-3.6×10^{12}	17.553	−223.50	-4.0×10^8
3Stage	−23.560	−40.084	−75.724	−22.861	−53.798	−199.18
3Stage (true λ)	−23.392	−38.930	−73.449	−22.260	−52.037	−190.57
Case 4: $\lambda = 1/2$, varying EVI						
QR	28.615	197.21	4562.8	−5.4690	−5.7396	2664.3
2Stage	−16.738	−56.502	10378.	−12.058	−47.251	2.1×10^5
3Stage	−16.347	−57.471	−669.89	−12.534	−50.538	−675.49
3Stage (true λ)	−19.682	−67.889	−824.30	−14.460	−56.746	−791.55
Case 5: $\lambda = 0$, varying EVI						
QR	37.747	56.769	59.097	15.016	32.656	269.83
2Stage	-5.1×10^4	-1.4×10^8	-1.5×10^{16}	−6.9535	−1545.6	-1.1×10^{11}
3Stage	−19.997	−34.041	−10.141	−14.737	−38.982	−184.99
3Stage (true λ)	−22.931	−41.986	−123.96	−15.408	−40.139	−190.74
Case 6: Fréchet, monotone varying EVI						
QR	1.9967	5.0835	41.914	−0.2840	0.5710	16.332
2Stage	−2.8044	−4.5649	3827.3	−1.8189	−4.3406	−25.209
3Stage	−2.4978	−4.3931	−12.692	−1.0843	−1.6962	−0.9360
Case 7: Fréchet, nonmonotone varying EVI						
QR	1.5309	9.5301	29.220	0.4664	1.6719	19.748
2Stage	−1.9793	−4.0033	−17.504	−1.1745	−2.9692	−17.012
3Stage	−3.0136	−5.2760	−13.916	−0.9322	−1.2815	7.7650

Table B3: Root Integrated Mean Squared Error (RIMSE) of estimators of $Q_Y(\tau_n \mid \mathbf{x})$ across sensitivity cases S1–S3, for $\tau_n \in \{0.99, 0.995, 0.999\}$ and $n \in \{500, 2000\}$. All cases replicate Case 4 with $\lambda = 1/2$, $\beta = (2,1,1)^\top$, varying only σ. Companion to Table 2.

	$n = 500$			$n = 2000$		
Method	$\tau_n = 0.99$	0.995	0.999	0.99	0.995	0.999
Case S1: $\gamma(\mathbf{x}) \in [0.45, 0.55]$*, near-constant EVI*						
QR	46.147	300.86	3526.6	14.105	42.021	955.84
2Stage	20.036	56.912	2826.7	29.086	3445.3	2.7×10^8
3Stage	21.572	50.475	419.49	11.988	29.215	234.07
3Stage (true λ)	19.336	43.327	278.60	11.568	27.650	203.88
Case S2: $\gamma(\mathbf{x}) \in [0.35, 0.65]$*, moderate EVI variation*						
QR	51.212	202.56	2.3×10^4	22.642	69.939	6709.4
2Stage	103.55	2928.0	7.7×10^6	21.753	103.18	7.6×10^5
3Stage	167.31	366.07	3058.8	20.992	59.917	1078.0
3Stage (true λ)	30.962	80.573	711.06	19.379	54.553	544.17
Case S3: $\gamma(\mathbf{x}) \in [0.25, 0.75]$*, large EVI variation (identical to Case 4)*						
QR	251.88	2320.1	6.1×10^4	56.642	166.88	3.6×10^4
2Stage	114.69	2215.9	3.9×10^6	58.751	1824.3	2.1×10^8
3Stage	80.807	259.10	7097.5	47.294	169.20	7597.4
3Stage (true λ)	62.811	183.08	2019.4	48.688	151.20	1825.0

Table B4: Integrated Bias (IBias) of estimators of $Q_Y(\tau_n \mid \mathbf{x})$ across sensitivity cases S1–S3, for $\tau_n \in \{0.99, 0.995, 0.999\}$ and $n \in \{500, 2000\}$. All cases replicate Case 4 with $\lambda = 1/2$, $\beta = (2,1,1)^\top$, varying only σ. Companion to Table 2.

	$n = 500$			$n = 2000$		
Method	$\tau_n = 0.99$	0.995	0.999	0.99	0.995	0.999
Case S1: $\gamma(\mathbf{x}) \in [0.45, 0.55]$*, near-constant EVI*						
QR	10.607	51.595	433.68	1.6115	8.1355	212.19
2Stage	−4.2956	−15.966	−138.41	−2.3617	−7.8241	2.7×10^5
3Stage	−5.4797	−17.902	−147.02	−2.7435	−12.486	−132.49
3Stage (true λ)	−5.4043	−18.319	−157.54	−2.6632	−12.609	−136.42
Case S2: $\gamma(\mathbf{x}) \in [0.35, 0.65]$*, moderate EVI variation*						
QR	10.031	43.493	1596.3	0.2437	5.3639	702.84
2Stage	−8.2963	−24.053	15223.	−4.6873	−20.167	756.59
3Stage	−8.0816	−27.333	−252.02	−5.0498	−22.022	−263.64
3Stage (true λ)	−9.6647	−31.782	−307.54	−5.3861	−23.285	−281.97
Case S3: $\gamma(\mathbf{x}) \in [0.25, 0.75]$*, large EVI variation (identical to Case 4)*						
QR	28.615	197.21	4562.8	−5.4690	−5.7396	2664.3
2Stage	−16.738	−56.502	10378.	−12.058	−47.251	2.1×10^5
3Stage	−16.347	−57.471	−669.89	−12.534	−50.538	−675.49
3Stage (true λ)	−19.682	−67.889	−824.30	−14.460	−56.746	−791.55

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